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Circulating Tumor DNA Methylation Is a Biomarker of Poor Recurrence-free Survival in Locally-Advanced Rectal Cancer

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Abstract

Background/Aim: The aim of this study was to investigate locus-specific circulating-tumor DNA (ctDNA) methylation for predicting long-term outcomes of locally-advanced rectal cancer (LARC) after resection.

Materials and Methods: In the present study, there were 50 patients without preoperative treatment and 35 patients with preoperative treatment. Methylation analyses for checkpoint with forkhead and ring finger domains (*CHFR*), sex-determining region Y-box transcription factor 11 (*SOX11*) and cysteine dioxygenase type 1 (*CDO1*) used DNA extracted from plasma ctDNA at the time of resection of the primary tumor with curative-intent surgery.

Results: Highly-methylated *SOX11* in ctDNA was found to be a biomarker of reduced recurrence-free survival (RFS) in LARC. In multivariate analysis, highly methylated *SOX11* was an independent prognostic factor for reduced RFS in the group without preoperative treatment.

Conclusion: The present study demonstrates that elevated ctDNA methylation of the *SOX11* gene is a biomarker for reduced RFS after curative-intent resection of LARC. Patients with high ctDNA methylation of the *SOX11* gene may not be optimal candidates for LARC resection. A prospective study is necessary to further validate *SOX11* ctDNA methylation as a biomarker for RFS of patients with LARC.

Keywords: Locally-advanced rectal cancer, LARC, SOX11, DNA methylation, liquid biopsy, circulating tumor DNA, ctDNA, prognostic biomarker, recurrence-free survival, preoperative treatment.

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Introduction

Colorectal cancer (CRC) is increasingly common and a major cause of mortality worldwide (1, 2). Approximately 30% of CRC cases are rectal cancer (1), which is an aggressive disease with poor long-term recurrence-free survival (RFS) and cancer-specific survival (3). Preoperative treatment (PT) for locally advanced rectal cancer (LARC) has improved postoperative outcomes (4, 5). In Western countries, PT is commonly used for resectable LARC, with subsequent total mesorectal excision (3, 6-8), but PT is less common in Japan (9-11). Regardless of the approach, LARC has a high local-recurrence rate of 5-10% after radical resection (12-16). Distant metastases occur at a rate of 25-40% and are mainly responsible for treatment failure (17, 18). This background suggests the need for improvement of the indication for resection in LARC to improve long-term outcomes.

Prognostic factors for LARC include imaging markers (extramural venous invasion, inter-rectal fascial involvement), blood markers [carcinoembryonic antigen (CEA), neutrophil-lymphocyte ratio], and pathological and molecular markers (tumor grade, tumor-infiltrating lymphocytes, circumferential resection margin) (19-21). CEA has been used as an indicator of tumor burden with considerable clinical benefits (22-25). Liquid biopsy based on sampling and analysis of blood and other non-solid biological tissue has more recently been introduced for disease monitoring of cancer and has the advantage of being non-invasive (26, 27). Advances in detection of circulating tumor DNA (ctDNA) have facilitated the introduction of liquid biopsy assays into clinical practice (27).

Hoffman and Diala discovered aberrant DNA methylation in cancer more than 40 years ago (28). Development and progression of CRC is influenced by altered DNA methylation (29, 30). Altered DNA methylation in tumor-suppressor genes is a prognostic factor in CRC and can also be used to evaluate susceptibility to anticancer agents (31, 32). Altered DNA methylation is a useful biomarker because CRC has many aberrantly methylated genes (32). In LARC, PT and subsequent curative-intent surgical resection of LARC can lead to improved outcomes, but the indication is uncertain (33). Thus, the aim of the present study was to investigate use of locus-specific ctDNA methylation analysis as a biomarker for predicting long-term outcomes of LARC after resection.

Materials and Methods

Study population. The subjects were patients who underwent resection of LARC with and without PT at Juntendo University Hospital, Japan, between 2011 and 2019. Methylation analyses used ctDNA extracted from patient plasma and DNA from corresponding frozen colorectal tumors at the time of resection of the primary tumor in surgery with curative intent. All procedures followed the ethical standards of institutional and national committees on human studies and the Helsinki Declaration of 1964 and revisions. Informed consent or the equivalent was obtained from all patients for inclusion in the study. Approval for the study was given by the Institutional Review Board at Juntendo University Hospital (IRB no. E23-0163).

Surgical strategies for rectal cancer. Resection of rectal cancer with lymph-node dissection at the root of the inferior mesenteric arteries and veins was performed with curative intent (34) using the principle of total mesorectal excision (35). Dissection was performed at least 2 cm distal of the cancer. A 2-cm mucosal margin above the dentate line permits preservation of the anal sphincter (36). Alternatively, abdominoperineal resection can be used. Open or laparoscopic surgery was chosen based on the tumor (site, progression) and patient factors (obesity, history of abdominal surgery). Some changes in indications occurred over the study period.

Preoperative treatment. Neoadjuvant chemoradiotherapy (NACRT) consisted of 45-50.4 Gy (1.8 Gy × 25-28 fractions) with oral 5-fluorouracil pro-drug administration. Neo-adjuvant chemotherapy (NAC) consisted of folinic acid/fluorouracil/oxaliplatin (FOLFOX, 6 cycles) or capecitabine–oxaliplatin (CAPOX, 4 cycles) with/without molecular-targeted agents. NACRT or NAC was selected based on age, performance status and clinical factors, and finally decided, for each case, based on a discussion between the physician and patient.

The histological criteria for assessment of the response to chemotherapy and radiotherapy (34) were: Grade 0 (no effect), no cancer-cell necrosis or degeneration in response to treatment; grade 1 (mild effect) minimal effect as cancer-cell necrosis or degeneration in <33% of the entire lesion, or mild effect as cancer-cell necrosis, degeneration, and/or lytic change in >33% but <67% of the entire lesion; grade 2 (moderate effect) prominent cancer-cell necrosis, degeneration, lytic change, and/or disappearance in >67% of the entire lesion, but viable cancer-cells remain; and grade 3 (marked effect) necrosis and/or lytic change throughout the entire lesion replaced by fibrosis with or without granulomatous change, with no viable cancer-cells.

Postoperative adjuvant chemotherapy. Postoperative adjuvant chemotherapy was recommended for all eligible stage III or high-risk stage II cases. A high-risk stage II case was defined as meeting at least one of the following criteria: T4, perforation/penetration, poorly-differentiated adenocarcinoma, mucinous carcinoma, and <12 examined lymph nodes (37). The postoperative adjuvant chemotherapy regimens changed over the study period because of the length of time. However, most patients received a 5-fluorouracil pro-drug orally or an oxaliplatin-based regimen intravenously for more than 6 months, starting 4 to 8 weeks after surgery.

Clinicopathological analyses. All specimens were examined as described elsewhere (34). Briefly, after resection of the rectum, the excised colorectal specimen was cut open by the surgeon. After formalin fixation, the specimen and lymph nodes were examined by a pathologist. Age, sex, venous invasion, lymphatic invasion, T classification, lymph-node metastasis, and preoperative serum CEA were recorded as clinicopathological factors. All cases were classified using the eighth edition of the Union for International Cancer Control classification (38).

Isolation of ctDNA from plasma and bisulfite conversion. Isolation of ctDNA from plasma samples (39, 40) was performed prior to surgery (after PT in the PT group), using 20 ml of blood collected in sodium heparin tubes (Becton Dickinson, Franklin Lakes, NJ, USA). Plasma was separated by centrifugation. The procedure, termed "Methylation on Beads", allows DNA extraction and bisulfite conversion in one tube with the use of silica super-magnetic beads (41), with a 1.5- to 5-fold better extraction efficiency than other techniques (42). The protocol (42) was optimized using 2.0 ml of plasma and 375 μl (800 units/ml) of proteinase K (New England Bio Labs, Ipswich, MA, USA), followed by digestion and then addition of 300 μl of isopropyl alcohol and 150 μl of MD1441 beads (Promega, Madison, WI, USA). The lysate was incubated and rotated for 10 minutes before addition of 5 ml of carrier RNA, after which incubation was continued for a further 5 minutes (42). DNA was subjected to bisulfite conversion using an EZ DNA Methylation™ Kit (Zymo Research, Irvine, CA, USA). The bisulfite conversion buffers used in the single-tube process were the same as those in the EZ DNA Methylation™ Kit. The final elution volume was adjusted to 100 μl. Finally, for desalting, the solution was purified using MicroSpin S-200 HR Columns (Cytiva, Tokyo, Japan) according to the manufacturer's instructions.

Genomic DNA extraction from cancer tissue and bisulfite conversion. Genomic DNA extraction was performed as described previously (43). Genomic DNA was purified from fresh frozen samples of resected primary tumor with an AllPrep DNA/RNA Mini Kit (Qiagen, Germantown, MD, USA). A sample (500 ng) of this DNA was bisulfite-converted using reagents in the EZ DNA Methylation Kit (Zymo Research) and samples were processed according to the manufacturer's instructions.

Quantitative methylation-specific polymerase chain reaction (qMSP). The extent of gene methylation was evaluated using ctDNA from preoperative blood samples. The level of methylation at CpG sites in the promoter region was assessed by qMSP analysis (39), using the bisulfite-modified

Table I. *Genes studied in the quantitative methylation-specific polymerase chain reaction.*

Gene	Gene function	Forward 5'-3'	Reverse 5'-3'	Probe 5'-3'	Annealing temperature (˚C)	Reference
CHFR	Maintenance of the antephase checkpoint that regulates cell-cycle entry into mitosis	GTT ATT TTC GTG A TT CGT AGG CGA C	CGA AAC CGA AAA TAA CCC GCG	\56-FAM\CGC TCG ACC\ZEN \ATC TTT AAT CCT AAC CAA ACG ACT TC\3IABkFQ\	60	42
SOX11	Regulation of embryonic development and cell fate	GGT AGG AGT TAC GAG TCG GAG AGA	ACT ACG ATC GCG ACA AAA AAA AC	6FAM-TCG GGT TGT TTC GAT CG-MGBNFQ	60	45
CDO1	Regulation of cysteine catabolism	CGT TTT TTT TCG TTT TAT TTT CGT CG	CCT CCG ACC CTT TTT ATC TAC G	TGT GGT TCG CGA CGT TGG GAC GT	60	47
ACTB	Housekeeping gene	TAG GGA GTA TAT AGG TTG GGG AAG TT	AAC ACA CAA TAA CAA ACA CAA ATT CAC	\56-FAM\TGT GGG GTG \ZEN\GTG ATG GAG GAG GTT TAG\3IABkFQ\	60	42

ACTB: β-actin; *CDO1*: cysteine dioxygenase type 1; *CHFR*: checkpoint with forkhead and ring finger domains; *SOX11*: sex-determining region Y-box transcription factor 11.

DNA as a template for fluorescence-based real-time PCR on an ABI StepOnePlus Real-Time PCR System (Applied BioSystems, Waltham, MA, USA). Promoter methylation of target genes (see below) was assessed in the bisulfite-modified DNA with 200 nM forward primer, 200 nM reverse primer and 80 nM probes. Cycling conditions were 95˚C for 5 minutes, followed by 55 cycles of 95˚C for 30 s, 60˚C for 30 s and 72˚C for 30 s. The master mix contained 16.6 mM $(NH_4)_2SO_4$, 67 mM Tris pH 8.8, 10 mM β-mercaptoethanol, 10 nM fluorescein, 0.166 mM of each deoxynucleotide triphosphate (dNTP) and 0. 04 U/μl Platinum Taq polymerase (ThermoFisher Scientific, Waltham, MA, USA) in a final reaction volume of 25 μl. Human female Jurkat genomic DNA treated with CpG methylase (M. *SssI*) (New England Bio Labs) served as a positive methylation control. Replicates for some samples produced no detectable methylation due to the low level of DNA methylation. Methylation was quantified using the relative methylation value (RMV), which was calculated as $2^{-\Delta\Delta Ct}$ for each methylation replicate by comparison with the mean Ct for β-actin (*ACTB*) (38). Replicates that were not detected were given a Ct of 100, creating a near-zero value for $2^{-\Delta\Delta Ct}$. The mean $2^{-\Delta\Delta Ct}$ value (RMV) was calculated as: mean $2^{-\Delta\Delta Ct}$ (RMV)=$(2^{-\Delta\Delta Ct\_replicate\_1} + 2^{-\Delta\Delta Ct\_replicate\_2} + 2^{-\Delta\Delta Ct\_replicate\_3})/3$ (39).

Three genes, namely checkpoint with forkhead and ring finger domains (*CHFR*), sex-determining region Y-box transcription factor 11 (*SOX11*), and cysteine dioxygenase type 1 (*CDO1*), were chosen for analysis, as discussed elsewhere. *CHFR* is a mitotic-checkpoint and tumor-suppressor gene that is inactivated mostly by promoter CpG island methylation, and *CHFR* methylation indicates a poor prognosis in cancer (43, 44). Aberrant DNA methylation of *SOX11* occurs in prostate, gastric and ovarian cancer and chronic lymphocytic leukemia (45). *SOX11* is a tumor-suppressor gene for which methylation is associated with poor survival in patients with these cancer types (45, 46). In carcinogenesis, abnormal *CDO1* regulation involves epigenetic changes in the promoter, and the extent of these changes is related to the progression and prognosis of cancer (47-49). Primers and probes for qMSP for the three genes are shown in Table I (43, 46, 48).

Statistical analysis. Discrete variables were compared using a Fisher exact probability test, and continuous variables were compared by the Mann−Whitney *U*-test for individual comparisons and Wilcoxon signed-rank test for paired comparisons. The cumulative-survival rate was calculated using the Kaplan−Meier method and univariate analyses

Table II. *Characteristics of the patients in the groups with and without preoperative treatment (PT).*

Clinicopathological factor		No-PT group n=50	PT group n=35	*p*-Value
Age, years	Median (range)	69 (39-91)	64 (43-86)	0.03
Sex, n (%)	Male	33 (66.0%)	27 (77.1%)	0.34
	Female	17 (34.0%)	8 (22.9%)	
Tumor location, n (%)	Ra	20 (40.0%)	9 (25.7%)	0.25
	Rb	30 (60.0%)	26 (74.3%)	
Serum CEA at primary tumor resection, ng/ml	Median (range)	2.7 (0.5-36.8)	2.5 (0.6-19.3)	0.79
Maximum diameter of primary tumor, mm	Median (range)	38.5 (18-105)	35 (0-160)	0.71
Predominant histological type of primary tumor, n (%)	Differentiated	44 (88.0%)	29 (82.9%)	0.54
	Undifferentiated[a]	6 (12.0%)	6 (17.1%)	
Undifferentiated component in primary tumor[a], n (%)	Absent	37 (74.0%)	27 (77.1%)	0.80
	Present	13 (26.0%)	8 (22.9%)	
T Classification, n (%)	pT/ypT 0, 1, 2	22 (44.0%)	10 (28.6%)	0.18
	pT/ypT 3, 4	28 (56.0%)	25 (71.4%)	
N Classification, n (%)	pN/ypN 0	32 (64.0%)	22 (62.9%)	>0.99
	pN/ypN 1, 2	18 (36.0%)	13 (37.1%)	
Lymphatic invasion (ly), n (%)	Absent	34 (68.0%)	30 (85.7%)	0.08
	Present	16 (32.0%)	5 (14.3%)	
Venous invasion (v), n (%)	Absent	33 (66.0%)	30 (85.7%)	0.048
	Present	17 (34.0%)	5 (14.3%)	
Histological criteria for assessment of the response to PT, n (%)	Grade 0, 1	N/A	19 (54.3%)	–
	Grade 2, 3	N/A	16 (4.3%)	

CEA: Carcinoembryonic antigen; N/A: not applicable; Ra: above the peritoneal reflection; Rb: below the peritoneal reflection. [a]Poorly-differentiated or mucinous adenocarcinoma or signet ring cell carcinoma.

were performed by a log-rank test. A Cox proportional-hazards regression model was used with hazard ratios (HR) and confidence intervals. Cut-offs for RMVs and clinicopathological factors for optimal prediction of long-term outcome (RFS) in each group were determined using a receiver operating characteristic curve, as the value that gave the largest area under the curve (AUC). Data were analyzed using JMP Pro 18 (SAS Institute Inc., Cary, NC, USA) with differences considered significant at $p \leq 0.05$. Values are expressed as a median (range).

Results

Patients. The study population comprised 85 patients who underwent curative-intent surgery for pathologically-confirmed stage I-III rectal cancer and had available preoperative plasma samples. There were 50 patients in the no-PT group and 35 patients in the PT group. PT included NACRT in 11 patients and NAC in 24 patients. Patient characteristics in the no-PT and PT groups are shown in Table II. Univariate analysis indicated that age and venous invasion differed significantly between no-PT and PT groups: patients in the no-PT group were significantly older (p=0.03) and more frequently had a tumor with venous invasion (p=0.048) compared to the PT group. The two groups had no significant difference in RFS (5-year-RFS: 77.4% in the no PT group and 83.5% in the PT group) (p=0.42).

Comparison of RMV for each gene between ctDNA and DNA extracted from cancer tissues. RMVs for each gene in ctDNA and DNA from cancer tissues were determined to identify potential correlation. RMVs in ctDNA and DNA from cancer tissues were not significantly correlated for *CHFR* (r=0.091, p=0.59); *SOX11* (r=0.224, p=0.18); and *CDO1* (r=0.125, p=0.46) in the no-PT group (Figure 1); nor for *CHFR* (r=0.312, p=0.17); *SOX11* (r=0.062, p=0.79); and *CDO1* (r=0.096, p=0.65) in the PT group (Figure 2).

Comparison of ctDNA RMVs for each gene according to clinicopathological factors. The optimal cut-off for the RMV

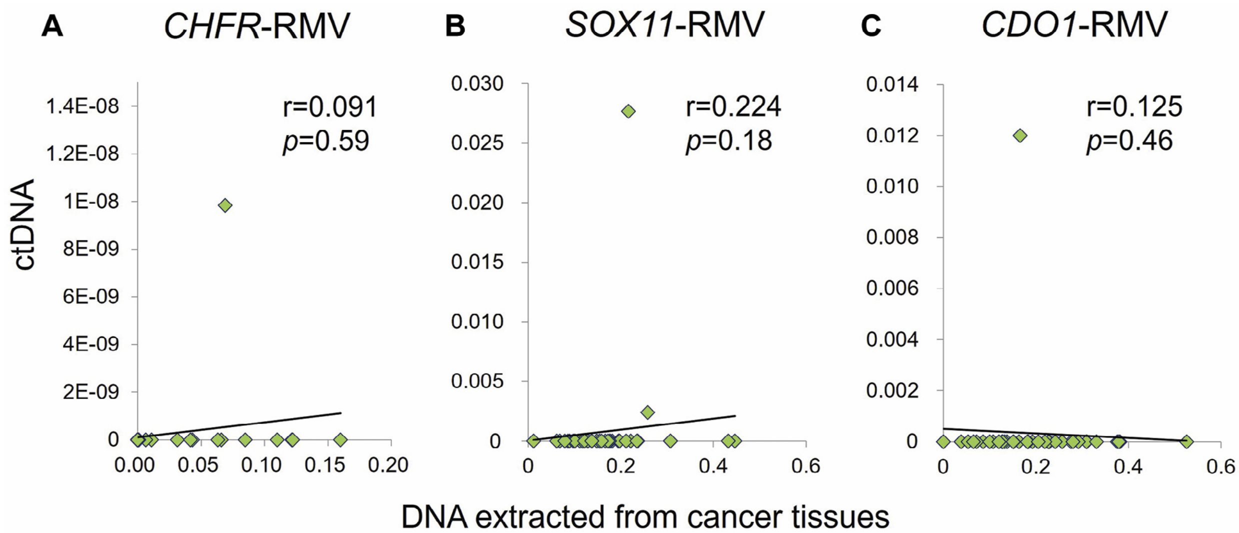

Figure 1. *Correlation of relative methylation values (RMVs) for (A) checkpoint with forkhead and ring finger domains (CHFR); (B) sex-determining region Y-box transcription factor 11 (SOX11); and (C) cysteine dioxygenase type 1 (CDO1); in circulating tumor DNA (ctDNA) and DNA extracted from cancer tissues in the group with no preoperative treatment. RMVs in ctDNA and DNA extracted from cancer tissues were not significantly correlated for CHFR (r=0.091, p=0.59); SOX11 (r=0.224, p=0.18); and CDO1 (r=0.125, p=0.46).*

for each gene corresponding to RFS was determined for each group. In the no-PT group, the RMV cut-off values giving the largest AUCs (in parentheses) were $1.017E^{-20}$ (0.557) for *CHFR*; $8.570E^{-14}$ (0.753) for *SOX11*; and $1.575E^{-09}$ (0.622) for *CDO1*. The patient characteristics of the no-PT group are shown in Table III. In univariate analysis, there was no significant difference in clinico-pathological factors between cases with high and low *CHFR*-RMV. *SOX11*-RMV^{high} cases had a significantly higher rate of N1/2 staging (*p*=0.02). *CDO1*-RMV^{high} cases had a trend towards a higher rate of lymphatic invasion (*p*=0.06). There was no significant difference between *CDO1*-RMV^{high} and *CDO1*-RMV^{low} cases.

In the PT group, the RMV cut-off values giving the largest AUCs (in parentheses) were $9.812E^{-21}$ (0.600) for *CHFR*; $6.755E^{-14}$ (0.613) for *SOX11*; and $7.006E^{-19}$ (0.567) for *CDO1*. The patient characteristics in the PT group are shown in Table IV. In univariate analysis, *CHFR*-RMV^{high} cases had a significantly higher rate of preoperative histological grade 2/3 (*p*=0.04) and a trend for a higher age (*p*=0.08). *SOX11*-RMV^{high} cases had a trend for a higher rate of lymphatic invasion (*p*=0.09), and *CDO1*-RMV^{high} cases had a trend towards a higher rate of N1/2 (*p*=0.08). There was no significant difference between RMV high and low cases for *SOX11* nor for *CDO1*.

RFS in high- and low-RMV cases for each gene. Long-term outcomes were evaluated based on RFS. In the no-PT group, there were no significant differences in RFS between *CHFR*-RMV^{high} and -RMV^{low} (*p*=0.12) and *CDO1*-RMV^{high} and -RMV^{low} (*p*=0.28) cases (Figure 3A and C). However, *SOX11*-RMV^{high} cases had a significantly worse RFS compared with *SOX11*-RMV^{low} cases (*p*<0.0001) (Figure 3B). In the PT group, there were no significant differences in RFS between *CHFR*-RMV^{high} and -RMV^{low} (*p*=0.20) and *CDO1*-RMV^{high} and -RMV^{low} (*p*=0.25) cases (Figure 4A and C). However, *SOX11*-RMV^{high} cases again had significantly worse RFS compared with *SOX11*-RMV^{low} cases (*p*=0.04) (Figure 4B).

In the no-PT group, in which *SOX11*-RMV^{high} cases had significantly worse RFS, a serum CEA level ≥5.0 ng/dl at the time of resection of the primary tumor was related to significantly worse RFS in univariate analysis (*p*=0.0004) (Table V). Multivariate analysis with these two variables revealed that the serum CEA level at the time of resection of the primary tumor (HR=6.14, 95% CI=1.62-23.32; *p*=0.008)

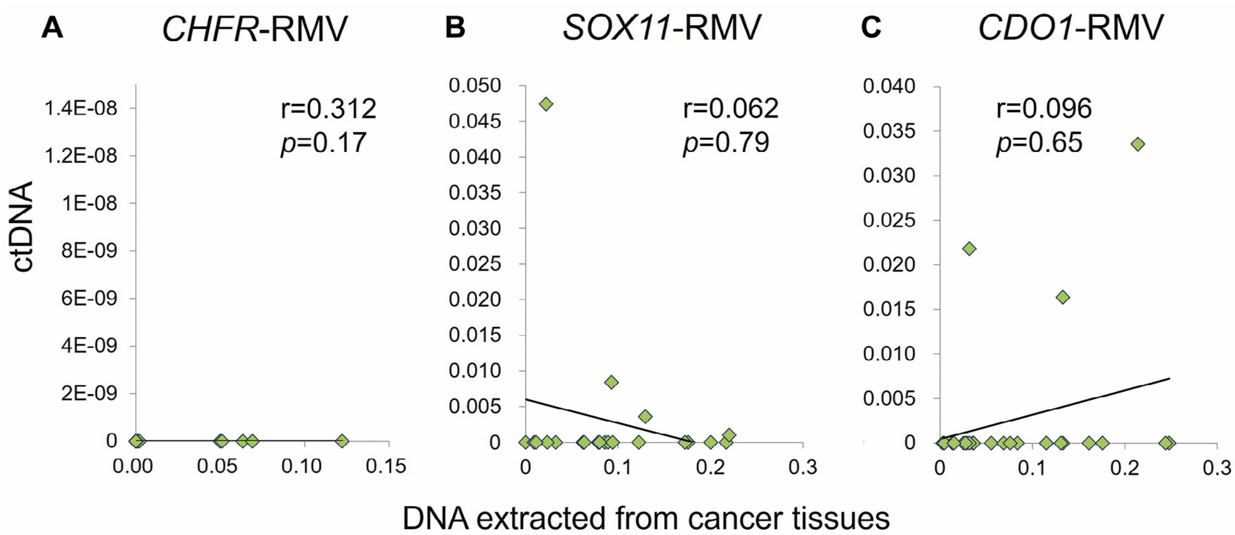

Figure 2. *Correlation of relative methylation values (RMVs) for (A) checkpoint with forkhead and ring finger domains (CHFR); (B) sex-determining region Y-box transcription factor 11 (SOX11); and (C) cysteine dioxygenase type 1 (CDO1); in circulating tumor DNA (ctDNA) and DNA extracted from cancer tissues in the group with preoperative treatment. RMVs in ctDNA and DNA extracted from cancer tissues were not significantly correlated for CHFR (r=0.312, p=0.17); SOX11 (r=0.062, p=0.79); and CDO1 (r=0.096, p=0.65).*

and *SOX11*-RMV (HR=8.59, 95%CI=2.33-31.71; *p*=0.001) were independent significant prognostic factors (Table V). In the PT group, in which *SOX11*-RMV^{high} cases also had significantly worse RFS, no other factor was significantly related to worse RFS in univariate analysis (Table VI). Therefore, multivariate analysis was not performed.

Discussion

In the present study, patients who had undergone curative-intent resection of LARC with and without PT, ctDNA *SOX11*-RMV^{high} cases had significantly worse RFS compared with ctDNA *SOX11*-RMV^{low} cases. Multivariate analysis revealed that ctDNA *SOX11*-RMV was an independent prognostic factor in patients without PT.

Studies of ctDNA in CRC using real-time PCR have identified increased methylation of the Septin 9 (*SEPT9)* gene (50, 51) in approximately 80% of patients with stage I-III CRC (52). Using methyl-BEAMing, which is based on methylated vimentin, Li *et al.* found an increased methylation rate in 59% of early-stage CRC cases (53).

In our previous study, we reported the prognostic utility of ctDNA methylation analysis in stage IV colorectal cancer (54). In previous studies (39, 40, 43-48), *CHFR, SOX11* and *CDO1* were chosen for investigation as genes that may predict long-term outcomes of LARC and be amenable to qMSP analysis using ctDNA from plasma. In the present study, high and low RMV for *SOX11* ctDNA was found to be useful to stratify RFS in no-PT cases, and in multivariate analysis, *SOX11*-RMV was an independent prognostic factor for RFS in these cases. *SOX11* is a neural transcription factor (55) that has been shown to have tumor-suppressor functions (56). Aberrant DNA methylation of *SOX11* is found in most endometrioid endometrial carcinomas and is correlated with clinicopathological factors in primary endometrial tumors (45). The methylation status of *SOX11* is also significantly associated with microsatellite instability and MutL homolog1 (*MLH1)* methylation in endometrial tumors (45). However, little is known about the function of *SOX11* in CRC, and further research is needed. In the present study, the clear stratification of RFS based on high and low RMV in *SOX11* ctDNA is presumably due to the sensitivity and specificity in the qMSP assay. However, there was no

Table III. *Comparisons of patient characteristics between cases with high and low relative methylation values (RMV) in the no-preoperative treatment group.*

Clinicopathological factor	Variable	ctDNA *CHFR*-RMV			ctDNA *SOX11*-RMV			ctDNA *CDO1*-RMV		
		Low (n=15)	High (n=35)	*p*-Value	Low (n=40)	High (n=10)	*p*-Value	Low (n=30)	High (n=20)	*p*-Value
Age, years	Median (range)	71 (43-91)	68 (39-86)	0.67	67.5 (39-91)	75 (61-82)	0.25	67.5 (39-86)	70.5 (52-91)	0.87
Sex, n (%)	Male	12 (80.0%)	21 (60.0%)	0.21	25 (62.5%)	8 (80.0%)	0.46	18 (60.0%)	15 (75.0%)	0.37
	Female	3 (20.0%)	14 (40.0%)		5 (37.5%)	2 (20.0%)		12 (40.0%)	5 (25.0%)	
Location of primary tumor, n (%)	Ra	4 (26.7%)	16 (45.7%)	0.35	17 (42.5%)	3 (30.0%)	0.72	12 (40.0%)	8 (40.0%)	>0.99
	Rb	11 (73.3%)	19 (54.3%)		23 (57.5%)	7 (70.0%)		18 (60.0%)	12 (60.0%)	
Serum CEA at primary tumor resection, ng/ml	Median (range)	2.9 (1.0-36.8)	2.6 (0.5-27.6)	0.37	2.7 (0.5-36.8)	3.1 (1.8-27.6)	0.10	2.5 (0.5-8.4)	2.9 (1.4-36.8)	0.12
Maximum diameter of primary tumor, mm	Median (range)	45 (18-86)	38 (20-105)	0.45	45 (23-70)	52.5 (25-135)	0.11	50 (23-120)	60 (35-135)	0.25
Predominant histological type of primary tumor, n (%)	Differentiated	14 (93.3%)	30 (85.7%)	0.65	36 (90.0%)	8 (80.0%)	0.59	27 (90.0%)	17 (85.0%)	0.67
	Undifferentiated[a]	1 (6.7%)	5 (14.3%)		4 (10.0%)	2 (20.0%)		3 (10.0%)	3 (15.0%)	
Undifferentiated component in primary tumor[a], n (%)	Absent	13 (86.7%)	24 (68.6%)	0.29	30 (75.0%)	7 (70.0%)	0.71	22 (73.3%)	15 (75.0%)	>0.99
	Present	2 (13.3%)	11 (31.4%)		10 (25.0%)	3 (30.0%)		8 (26.7%)	5 (25.0%)	
T Classification, n (%)	T1, 2	7 (46.7%)	15 (42.9%)	>0.99	20 (50.0%)	2 (20.0%)	0.15	15 (50.0%)	7 (35.0%)	0.39
	T3, 4	8 (53.3%)	20 (57.1%)		20 (50.0%)	8 (80.0%)		15 (50.0%)	13 (65.0%)	
N Classification, n (%)	N0	8 (53.3%)	24 (68.6%)	0.35	29 (72.5%)	3 (30.0%)	**0.02**	22 (73.3%)	10 (50.0%)	0.13
	N1, 2	7 (46.7%)	11 (31.4%)		11 (27.5%)	7 (70.0%)		8 (26.7%)	10 (50.0%)	
Lymphatic invasion) (ly), n (%)	Absent	12 (80.0%)	22 (62.9%)	0.33	27 (67.5%)	7 (70.0%)	>0.99	17 (56.7%)	17 (85.0%)	0.06
	Present	3 (20.0%)	13 (37.1%)		13 (32.5%)	3 (30.0%)		13 (43.3%)	3 (15.0%)	
Venous invasion (v), n (%)	Absent	11 (73.3%)	22 (62.9%)	0.53	28 (70.0%)	5 (50.0%)	0.28	18 (60.0%)	15 (75.0%)	0.37
	Present	4 (26.7%)	13 (37.1%)		12 (30.0%)	5 (50.0%)		12 (40.0%)	5 (25.0%)	

CDO1: Cysteine dioxygenase type 1; CEA: carcinoembryonic antigen; *CHFR*: checkpoint with forkhead and ring finger domains; Ra: above the peritoneal reflection; Rb: below the peritoneal reflection; *SOX11*: sex-determining region Y-box transcription factor 11. [a]Poorly-differentiated or mucinous adenocarcinoma or signet ring cell carcinoma. Statistically-significant *p*-values are shown in bold.

Table IV. *Comparisons of patient characteristics between cases with high and low relative methylation values (RMV) in the preoperative treatment (PT) group.*

Clinicopathological factor	Variable	ctDNA *CHFR*-RMV			ctDNA *SOX11*-RMV			ctDNA *CDO1*-RMV		
		Low (n=15)	High (n=35)	*p*-Value	Low (n=40)	High (n=10)	*p*-Value	Low (n=30)	High (n=20)	*p*-Value
Age, years	Median (range)	60.5 (43-83)	67 (48-86)	0.08	65 (43-86)	60 (55-76)	0.45	66 (43-86)	63 (48-83)	0.60
Sex, n (%)	Male	15 (83.3%)	12 (70.6%)	0.44	20 (76.9%)	7 (77.8%)	>0.99	13 (81.3%)	14 (73.7%)	0.70
	Female	3 (16.7%)	5 (29.4%)		6 (23.1%)	2 (22.2%)		3 (18.8%)	5 (26.3%)	
Location of primary tumor, n (%)	Ra	5 (27.8%)	4 (23.5%)	>0.99	6 (23.1%)	3 (33.3%)	0.66	2 (12.5%)	7 (36.8%)	0.14
	Rb	13 (72.2%)	13 (76.5%)		20 (76.9%)	6 (66.7%)		14 (87.5%)	12 (63.2%)	
Serum CEA at primary tumor resection, ng/ml	Median (range)	2.3 (0.9-12.5)	2.9 (0.6-19.3)	0.10	2.6 (0.6-13.3)	2.5 (1.4-19.3)	0.91	3.0 (0.6-12.5)	2.5 (0.9-19.3)	0.53
Maximum diameter of primary tumor, mm	Median (range)	35 (13-65)	40 (0-160)	0.80	35 (0-105)	40 (20-160)	0.26	35 (20-105)	35 (0-160)	0.25
Predominant histological type of primary tumor, n (%)	Differentiated	15 (83.3%)	14 (82.4%)	>0.99	21 (80.8%)	8 (88.9%)	>0.99	12 (75.0%)	17 (89.5%)	0.38
	Undifferentiated[a]	3 (16.7%)	3 (17.7%)		5 (19.2%)	1 (11.1%)		4 (25.0%)	2 (10.5%)	
Undifferentiated component in primary tumor[a], n (%)	Absent	14 (77.8%)	13 (76.5%)	>0.99	19 (73.1%)	8 (88.9%)	0.65	10 (62.5%)	17 (89.5%)	0.11
	Present	4 (22.2%)	4 (23.5%)		7 (26.9%)	1 (11.1%)		6 (37.5%)	2 (10.5%)	
T Classification, n (%)	ypT0, 1, 2	6 (33.3%)	4 (23.5%)	0.71	8 (30.8%)	2 (22.2%)	>0.99	5 (31.3%)	5 (26.3%)	>0.99
	ypT3, 4	12 (66.7%)	13 (76.5%)		18 (69.2%)	7 (77.8%)		11 (68.8%)	14 (73.7%)	
N Classification, n (%)	ypN0	11 (61.1%)	11 (64.7%)	>0.99	17 (65.4%)	5 (55.6%)	0.70	13 (81.3%)	9 (47.4%)	0.08
	ypN1, 2	7 (38.9%)	6 (35.3%)		9 (34.6%)	4 (44.4%)		3 (18.8%)	10 (52.6%)	
Lymphatic invasion (ly), n (%)	Absent	16 (88.9%)	14 (82.4%)	0.66	24 (92.3%)	6 (66.7%)	0.09	14 (87.5%)	16 (84.2%)	>0.99
	Present	2 (11.1%)	3 (17.6%)		2 (7.7%)	3 (33.3%)		2 (12.5%)	3 (15.8%)	
Venous invasion (v), n (%)	Absent	15 (83.3%)	15 (88.2%)	>0.99	22 (84.6%)	8 (88.9%)	>0.99	14 (87.5%)	16 (84.2%)	>0.99
	Present	3 (16.7%)	2 (11.8%)		4 (15.4%)	1 (11.1%)		2 (12.5%)	3 (15.8%)	
Histological criteria for assessment of the response to PT, n (%)	Grade 0, 1	13 (72.2%)	6 (35.3%)	**0.04**	12 (46.2%)	7 (77.8%)	0.14	9 (56.3%)	10 (52.6%)	>0.99
	Grade 2, 3	5 (27.8%)	11 (64.7%)		14 (53.9%)	2 (22.2%)		7 (43.8%)	9 (47.4%)	

CDO1: Cysteine dioxygenase type 1; CEA: carcinoembryonic antigen; *CHFR*: checkpoint with forkhead and ring finger domains; Ra: above the peritoneal reflection; Rb: below the peritoneal reflection; *SOX11*: sex-determining region Y-box transcription factor 11. [a]Poorly-differentiated or mucinous adenocarcinoma or signet ring cell carcinoma. Statistically-significant *p*-values are shown in bold.

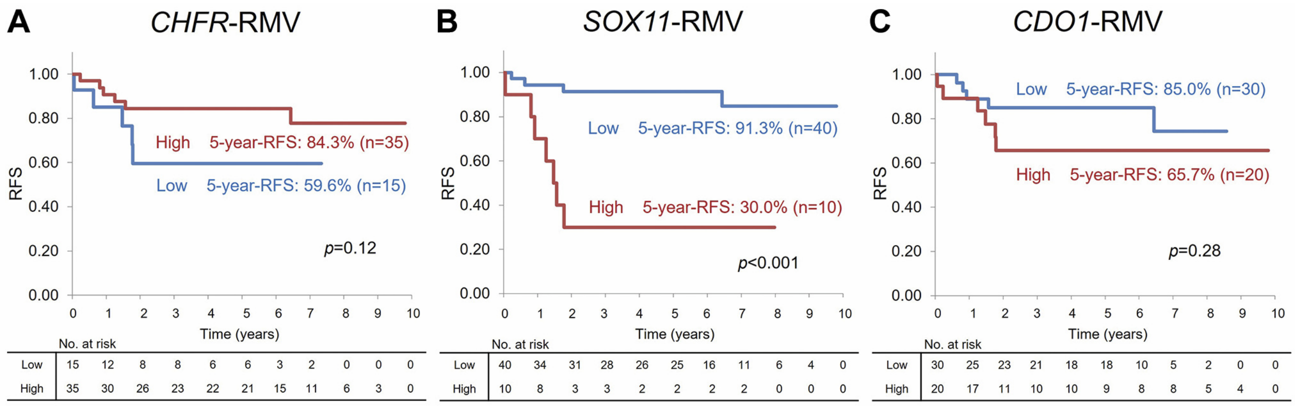

Figure 3. *Recurrence-free survival (RFS) in the group with no preoperative treatment according to relative methylation value (RMV) for (A) checkpoint with forkhead and ring finger domains (CHFR); (B) sex-determining region Y-box transcription factor 11 (SOX11); and (C) cysteine dioxygenase type 1 (CDO1) in circulating tumor DNA (ctDNA). Cases with SOX11-RMV^{high} had significantly worse RFS compared with SOX11-RMV^{low} cases (p<0.0001). However, there were no significant differences in RFS between CHFR-RMV^{high} and -RMV^{low} (p=0.12) and CDO1-RMV^{high} and -RMV^{low} (p=0.28) cases.*

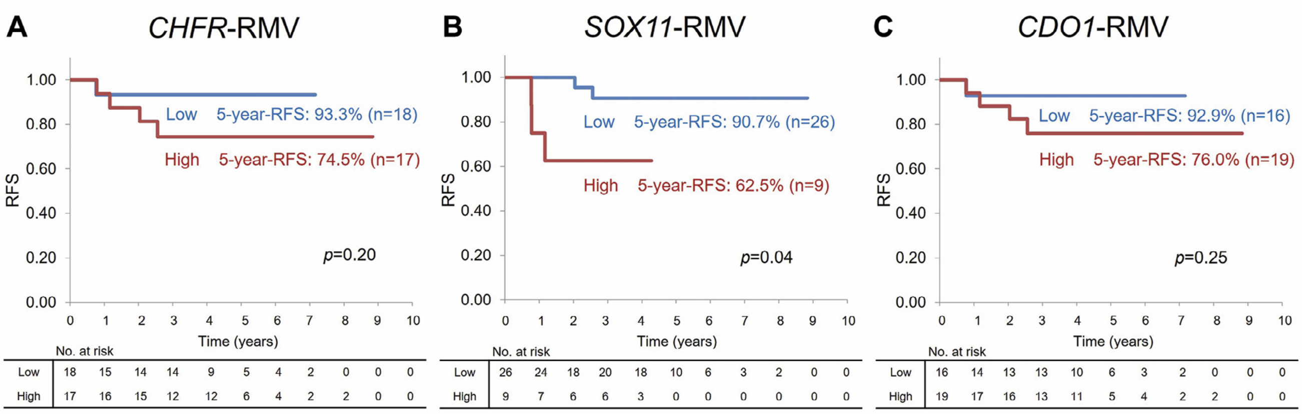

Figure 4. *Recurrence-free survival (RFS) in the group with preoperative treatment according to relative methylation value (RMV) for (A) checkpoint with forkhead and ring finger domains (CHFR); (B) sex-determining region Y-box transcription factor 11 (SOX11); and (C) cysteine dioxygenase type 1 (CDO1) in circulating tumor DNA (ctDNA). Cases with SOX11-RMV^{high} had significantly worse RFS compared with SOX11-RMV^{low} cases (p=0.04). However, there were no significant differences in RFS between CHFR-RMV^{high} and -RMV^{low} (p=0.20) and CDO1-RMV^{high} and -RMV^{low} (p=0.25) cases.*

significant correlation of *CHFR*-, *SOX11*- and *CDO1*-RMVs in plasma ctDNA and DNA from cancer tissues, and further examination of these results is required.

For *SOX11*, the RMVs were relatively higher than those for *CHFR* and *CDO1*. Thus, the sensitivity and specificity of RMVs for *SOX11* in ctDNA seemed to be appropriate to stratify the prognosis of LARC cases with and without PT, in which ctDNA was more likely to be shed into the bloodstream.

Whole-exome sequencing of ctDNA from postsurgical plasma samples was recently shown to be useful to identify patients at increased risk of recurrence who were likely to benefit from adjuvant chemotherapy in stage II-IV resectable CRC (57). Postsurgical ctDNA positivity might reflect the presence of minimal residual disease (MRD). It has been suggested that ctDNA status during treatment or postoperatively may permit detection of MRD in the adjuvant setting (58). Moreover, integration of genomic and

Table V. *Univariate and multivariate analyses related to the recurrence-free survival (RFS) in the no-preoperative treatment group.*

Clinicopathological factor	Variable	n	5-Year RFS	Univariate analysis *p*-Value	Multivariate analysis		
					HR	95% CI	*p*-Value
Age	<70 Years	26	82.9%	0.45	–	–	–
	≥70 Years	24	71.1%				
Sex	Male	33	72.3%	0.21	–	–	–
	Female	17	87.1%				
Location of primary tumor	Ra	20	79.7%	0.78	–	–	–
	Rb	30	76.0%				
Serum CEA level at resection of primary tumor	<5.0 ng/dl	40	88.2%	**0.0004**	6.14	1.62-23.32	**0.008**
	≥5.0 ng/dl	10	40.0%				
Predominant histological type of primary tumor	Differentiated	44	79.5%	0.31	–	–	–
	Undifferentiated[a]	6	60.0%				
Undifferentiated component in primary tumora	Absent	37	81.9%	0.07	–	–	–
	Present	13	63.6%				
T Classification	T1, 2	22	88.8%	0.29	–	–	–
	T3, 4	28	69.3%				
N Classification	N0	32	85.9%	0.12	–	–	–
	N1, 2	18	62.8%				
Lymphatic invasion (ly)	Absent	34	76.7%	0.69	–	–	–
	Present	16	78.6%				
Venous invasion (v)	Absent	33	82.3%	0.17	–	–	–
	Present	17	69.0%				
ctDNA *SOX11*-RMV	Low	40	91.3%	**<0.0001**	8.59	2.33-31.71	**0.001**
	High	10	30.0%				

CEA: Carcinoembryonic antigen; CI: confidence interval; HR: hazard ratio; Ra: above the peritoneal reflection; Rb: below the peritoneal reflection; *SOX11*: sex-determining region Y-box transcription factor 11. [a]Poorly-differentiated or mucinous adenocarcinoma or signet ring cell carcinoma. Statistically-significant *p*-values are shown in bold.

epigenomic assessment of ctDNA in plasma samples at about 4 weeks after surgery can improve the sensitivity of MRD detection (59). For these reasons, use of plasma ctDNA after surgery for MRD detection is of interest; however, no such samples after surgery were available in the present study. Since LARC resection places a huge surgical stress on patients compared to patients with colon cancer, use of *SOX11*-RMV in ctDNA as an independent prognostic factor for RFS in the no-PT and PT groups may be beneficial for determining the validity of surgical intervention.

There are several limitations to this study, including the small number of patients and the lack of data for samples after surgery. In addition, only one gene, *SOX11*, was identified as an independent prognostic factor. Three genes were chosen based on a pilot study, but others may also predict long-term outcomes using high- and low-RMV determination, and further work is needed to identify such genes.

Conclusion

The present study shows the utility of RMV analyses of the *SOX11* gene in ctDNA as a biomarker for prediction of long-term outcomes after resection of LARC. A prospective study is needed to further validate *SOX11*-gene RMV in ctDNA as a predictive biomarker.

Conflicts of Interest

The Authors declare that they have no conflicts of interest in relation to this study.

Authors' Contributions

K. Sugimoto developed the concept of the study and drafted the article. K. Sugimoto, T. Irie and H. Momose

Table VI. *Univariate analyses related to the recurrence-free survival (RFS) in the group with preoperative treatment.*

Clinicopathological factor	Variable	n	5-Year RFS	Univariate analysis *p*-Value
Age	<70 Years	26	81.9%	0.76
	≥70 Years	9	87.5%	
Sex	Male	27	77.5%	0.16
	Female	8	100%	
Location of primary tumor	Ra	9	100%	0.17
	Rb	26	77.8%	
Serum CEA level at resection of primary tumor	<5.0 ng/dl	29	80.2%	0.30
	≥5.0 ng/dl	6	100%	
Predominant histological type of primary tumor	Differentiated	29	80.2%	0.30
	Undifferentiated[a]	6	100%	
Undifferentiated component in primary tumor[a]	Absent	27	78.5%	0.20
	Present	8	100%	
T Classification	ypT0, 1, 2	10	77.8%	0.56
	ypT3, 4	25	85.9%	
N Classification	N0	22	89.7%	0.23
	N1, 2	13	72.7%	
Lymphatic invasion (ly)	Absent	30	85.3%	0.35
	Present	5	66.7%	
Venous invasion (v)	Absent	30	84.4%	0.89
	Present	5	80.0%	
Histological criteria for assessment of the response to PT	Grade 0, 1	19	80.0%	0.50
	Grade 2, 3	16	87.1%	
ctDNA *SOX11*-RMV	Low	26	90.7%	**0.04**
	High	9	62.5%	

CEA: Carcinoembryonic antigen; CI: confidence interval; HR: hazard ratio; Ra: above the peritoneal reflection; Rb: below the peritoneal reflection; *SOX11*: sex-determining region Y-box transcription factor 11. [a]Poorly-differentiated or mucinous adenocarcinoma or signet ring cell carcinoma. Statistically-significant *p*-values are shown in bold.

performed genomic DNA extraction, bisulfite treatment and qMSP. K. Sugimoto, T. Irie and H. Momose performed the patient survey and analyzed data. S. Kochi, M. Toake, Y. Tsuchiya, R. Tsukamoto, K. Honjo, S. Ishiyama, M. Takahashi and K. Sakamoto recruited patients. K. Sugimoto and K. Sakamoto obtained IRB approval for the protocol of the study. R.M. Hoffman revised the article. The article has been approved by all Authors.

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